Student activity

From the map given above, name the plates causing the formation of fold mountains given below.

S. No	Fold Mountain	Two Converging Plates
1.	The Andes	→ ←
2.	The Rockies	→ ←
3.	The Atlas	→ ←
4.	The Himalayas	→ ←
5.	The Alps	→ ←

5. When one side of the fold is pushed so much that it lies positioned over the other, such a fold is called **recumbent fold**.

When plates converge, the weak rocks and sediments lying between two plates get squeezed and folded. Parallel folds form long chains of fold mountain ranges with high peaks. The fold mountains are characterised by peaks and valleys. The tops of anticlines become the peaks and synclines become the valleys. Intermontane plateaus (plateau surrounded by the mountain ranges all sides) may be found between the high ranges. Example, Tibet.

3.8 Fault

A fault is a break in earth's crust where blocks of rock crust slide past each other.

Usually it occurs along plate boundaries, where the forces of plate motion compress, pull or shear the crust that breaks the crust. Energy release associated with rapid movement on active faults is the cause of most earthquakes.

The **fault plane** is the flat surface along which broken blocks of rock slide past one another. A **fault dip** is an angle between the fault plane and horizontal plane. Up thrown side represents the uppermost block of a fault. Down thrown side represents the lowermost block of a fault. Sometimes it becomes difficult to find out, which block has really moved along the fault plane. Hanging wall is the upper wall of a fault. Foot wall represents the lower wall of a fault.

A **fault scarp** is the steep wall like slope caused by faulting of the crustal rocks. Sometimes the fault scrap is so steep that it resembles a cliff.

3.8.1 Types of Faults

Based on how plates move about, the fault can be divided into as follow:

3.8.1.1 Normal Fault

Vertical displacement of the crust is called a normal fault. The normal fault is caused by tensional forces where plates diverge. One block lies above the other (hanging wall). The other block lies below the fault (footwall).

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